

MAE 170 – Discussion Session 1

Introduction to Control Systems

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Office Hours

TA Contact

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Come with questions - I'm here to help!

How Do You Get a System to Do What You Want?

Control System provides the mathematical tools to answer this question.

Self-Driving Car

Navigate city streets autonomously

Building HVAC

Manage temperature efficiently

Insulin Pump

Regulate blood glucose levels

Anatomy of a Dynamical System

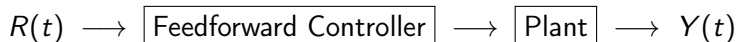
System Variables

Variable	Symbol	Description
Control inputs	$u(t)$	Intentional: steering, brakes, accelerator
Disturbances	$d(t)$	Unintentional: wind, road bumps
System state	$x(t)$	Evolves over time with inputs & dynamics

Inputs interact with internal dynamics to produce system state $x(t)$.

Open-Loop Control (Feedforward)

Signal flows forward only — NO feedback loop.



Strengths ✓

- Simple & inexpensive to implement
- No stability risk from feedback
- Effective when model is accurate

Limitations ×

- Cannot correct for disturbances
- Requires very accurate system model
- Errors accumulate over time

Closed-Loop Control (Feedback)

$$R(s) \longrightarrow \Sigma \longrightarrow \boxed{C(s)} \longrightarrow \boxed{G(s)} \longrightarrow Y(s)$$

Key Insight

Feedback is a **self-correcting mechanism**: if the system drifts due to disturbances or model errors, the controller detects the error and compensates.

Important

More powerful than feedforward but also more dangerous. Feedback *changes the dynamics* of the system, which can affect stability.

Ensure your design works before deploying it.

Bode Diagram

Frequency-domain: gain and phase vs. frequency

Nyquist Diagram

Stability analysis using open-loop frequency response

MATLAB / Simulink

Simulate full system before hardware testing

Symbolic Math Toolbox

What it does

- Performs *exact* symbolic math — no floating-point rounding
- Handles variables s, t, a, ω as mathematical symbols
- Computes Laplace transforms analytically (matches Table D.1)
- Does partial fraction decomposition symbolically

```
% Check toolbox availability (returns 1 if available)
license('test','symbolic_toolbox')

% Declare symbolic variables
syms s t a omega

% Now s, t, a, omega are symbols (not numbers!)
```

MATLAB Function Reference — Transforms

`laplace(f, t, s)`

Computes $\mathcal{L}\{f(t)\}$. Returns $F(s)$ as a symbolic expression.

```
laplace(exp(-a*t), t, s)  
% -> 1/(a+s)
```

`ilaplace(F, s, t)`

Computes $\mathcal{L}^{-1}\{F(s)\}$. Returns $f(t)$ as a symbolic expression.

```
ilaplace(1/(s+3), s, t)  
% -> exp(-3*t)
```

Table D.1 Verification Examples

Entry	$f(t)$	MATLAB	$F(s)$
4	e^{-at}	<code>laplace(f,t,s)</code>	$\frac{1}{s+a}$
6	$1 - e^{-at}$	<code>laplace(f,t,s)</code>	$\frac{a}{s(s+a)}$
16	$e^{-at} \sin(\omega t)$	<code>laplace(f,t,s)</code>	$\frac{\omega}{(s+a)^2 + \omega^2}$
17	$e^{-at} \cos(\omega t)$	<code>laplace(f,t,s)</code>	$\frac{s+a}{(s+a)^2 + \omega^2}$

MATLAB Function Reference — syms & diff

syms

Declares symbolic variables. Must be called before any symbolic computation.

```
syms s t a omega  
% Creates symbolic vars
```

diff(f, t, n)

Differentiates f w.r.t. t . Optional n sets order (default $n = 1$).

```
diff(exp(-a*t), t, 2)  
% -> a^2*exp(-a*t)
```

Differentiation Property Verification

```
syms f(t)  
Df = diff(f, t);  
laplace(Df, t, s)           % -> s*laplace(f,t,s) - f(0)  
% Concrete: f = e^(-at)  
f_ex = exp(-a*t);  
laplace(diff(f_ex,t), t, s)  
% -> -a/(s+a) = s*[1/(s+a)] - 1  checkmark
```

MATLAB Function Reference — Partial Fractions

`partfrac(F, s)`

Decomposes a rational $F(s)$ into partial fractions. Each term maps directly to a row in Table D.1 — read off $f(t)$ by inspection.

```
F2 = (2*s + 5) / ((s+1)*(s+2));  
partfrac(F2, s)  
% -> 3/(s+1) - 1/(s+2)  
ilaplace(F2, s, t)  
% -> 3*exp(-t) - exp(-2*t)
```

Inverse Laplace Workflow

$F(s)$ Rational $\xrightarrow{\text{partfrac}}$ Decompose $\xrightarrow{\text{Table D.1}}$ Match terms $\longrightarrow f(t)$

Example — Find $f(t)$

Given:

$$F(s) = \frac{s+3}{s^2+2s+5}$$

Hint: Complete the square $\Rightarrow s^2 + 2s + 5 = (s+1)^2 + 4 = (s+1)^2 + 2^2$

This is a **damped cosine** — Table D.1 Entry 17: $\frac{s+a}{(s+a)^2 + \omega^2}$

```
syms s t
F3 = (s+3) / (s^2 + 2*s + 5);
ilaplace(F3, s, t) % Verify your hand answer!
```

Expected Answer

$$f(t) = e^{-t}(\cos 2t + \sin 2t)$$

Complex Numbers — Rectangular Form

$$c = a + jb$$

Real part a , imaginary part b .
Use $1i$ or $1j$ in MATLAB (not bare i/j).

```
z = 3 + 4j;  
real(z), imag(z) % 3 and  
4
```

$$\text{Conjugate } c^* = a - jb$$

Flips sign of imaginary part.
Key identity:
 $c \cdot c^* = |c|^2 = a^2 + b^2$.

```
z * conj(z)  
% (3+4j)*(3-4j) = 25
```

Key Properties — Rectangular Form

Re/Im	<code>real(z)</code> , <code>imag(z)</code>	$z = a + jb$	a, b
Conjugate	<code>conj(z)</code>	$z = a + 1j \cdot b$	$a - jb$
Magnitude	<code>abs(z)</code>	$\text{abs}(z)^2$	$\sqrt{a^2 + b^2}$
Arithmetic	$z_1 + z_2$, $z_1 * z_2$	$(a_1 + a_2) + (b_1 + b_2)j$	add parts

Complex Numbers — Polar Form

$$c = r e^{j\varphi}$$

Magnitude $r = |c|$, phase $\varphi = \angle c$.
`abs()` returns r ; `angle()` returns $\varphi \in (-\pi, \pi]$.

```
abs(z), angle(z)  
% 5 and 0.9273 rad
```

Polar $\leftrightarrow$ Rectangular

$$a = r \cos \varphi, \quad b = r \sin \varphi.$$

Pole locations $s = -a \pm j\omega$ are complex — magnitude and angle define damping and frequency.

Key Properties — Polar Form

Magnitude	<code>abs(z)</code>	$r = c $	$\sqrt{a^2 + b^2}$
Phase	<code>angle(z)</code>	$\varphi = \angle c$	<code>atan2(b, a)</code>
Conjugate	<code>conj(z)</code>	$re^{-j\varphi}$	same r , flipped φ
Multiply	<code>z1*z2</code>	$r_1 r_2 e^{j(\varphi_1 + \varphi_2)}$	magnitudes multiply

Phase Angle — $\text{atan2}(b, a)$ Reference Table

The phase of $c = a + bj$ is $\varphi = \text{atan2}(b, a)$ — accounts for the correct quadrant.

Condition	Formula for φ	Range
$a > 0, b \geq 0$ (Q I)	$\arctan(b/a)$	0 to 90
$a > 0, b < 0$ (Q IV)	$\arctan(b/a)$	-90 to 0
$a < 0, b \geq 0$ (Q II)	$\arctan(b/a) + 180$	90 to 180
$a < 0, b < 0$ (Q III)	$\arctan(b/a) - 180$	-180 to -90
$a = 0, b > 0$	+90	+90
$a = 0, b < 0$	-90	-90
$a = 0, b = 0$	Undefined	—

Shortcut

When $a > 0$: $\varphi = \arctan(b/a)$ directly — no quadrant correction needed.

Example 1 — Complex Division

Find: $c = \frac{1+2j}{3+j}$ Method: multiply by conjugate of denominator.

- ① Conjugate of denominator: $(3+j)^* = (3-j)$
- ② Multiply: $c = \frac{(1+2j)(3-j)}{(3+j)(3-j)}$
- ③ Numerator: $(1+2j)(3-j) = 3-j+6j-2j^2 = 5+5j$
- ④ Denominator: $(3+j)(3-j) = 9+1 = 10$
- ⑤ Divide: $c = \frac{5+5j}{10} = 0.5 + 0.5j$

Answer

$$c = 0.5 + 0.5j, \quad |c| = \sqrt{0.5} \approx 0.707, \quad \angle c = 45$$

Example 2 — Find the Angle (Phase)

Find the angle of: $c = \frac{-j \cdot (2 + 2j)}{\sqrt{3} + j}$ Key:

$$\angle(A \cdot B/C) = \angle A + \angle B - \angle C$$

$$\angle(-j) \quad a = 0, b = -1 \Rightarrow -90$$

$$\angle(2 + 2j) \quad a = 2 > 0, b = 2 \Rightarrow \arctan(1) = 45$$

$$\angle(\sqrt{3} + j) \quad a = \sqrt{3} > 0, b = 1 \Rightarrow \arctan(1/\sqrt{3}) = 30$$

Total Angle

$$\angle c = -90 + 45 - 30 = -75 = -\frac{5\pi}{12} \text{ rad}$$

Example 3 — Phase of a Complex Number

Given: $C = -1 - 2j$ Find $\angle C$. $a < 0, b < 0 \Rightarrow$ Quadrant III.

- ① Identify: $a = -1, b = -2$
- ② Quadrant III $\Rightarrow \varphi = \arctan(b/a) - 180$
- ③ $\arctan(-2/-1) = \arctan(2) \approx 63.43$
- ④ Apply correction: $\varphi = 63.43 - 180 = -116.57$

Answer

$$\angle C = \arctan(2) - 180 \approx -116.57 \approx -2.034 \text{ rad} \quad (\text{Quadrant III})$$